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### Battery pack

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#### Abstract

A configuration is provided so as to enable efficient fixing of cell groups to a housing of a battery pack. A housing of a battery pack has inside thereof a partition wall which extends in an extending direction and projects in a projecting direction. A cell group has a plurality of battery cells side-by-side in the extending direction one a side of the partition wall. Paste is filled into a gap between the partition wall and the cell group. The partition wall has an injection channel. The injection channel has an injection port for injecting the paste, and an inlet which communicates with the injection port and opens to the gap.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC         |
|--------------|-------------|---------------|----------|-------------|
| 2019/0181404 | 12/2018     | Seki          | N/A      | H01M 50/507 |
| 2023/0057595 | 12/2022     | Vogel         | N/A      | H01M 10/653 |

### FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 2019106264 | 12/2018          | JP      | N/A |

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## Background/Summary

(1) This application is based on and claims the benefit of priority from Chinese Patent Application No. CN202210097849.X, filed on 27 Jan. 2022, the content of which is incorporated herein by reference.

### BACKGROUND OF THE INVENTION

#### Field of the Invention

(2) The present invention relates to a battery pack having a plurality of battery cells.

#### Related Art

(3) In recent years, from the viewpoint of decreasing the emission of carbon dioxide, reducing the negative impact on the global environment, etc., the popularization of electric vehicles such as EV and HEV has been progressing. For the battery pack equipped to an electric vehicle or the like, technology for expanding the occupied area by battery cells within the battery pack, and thus equipping many battery cells has been necessitated.

(4) More specifically, conventionally, a battery module has been formed by combining a plurality of battery cells, and further, a battery pack has been formed by mounting a plurality of these battery modules in a housing. Such a battery module has a strong structure due to accomplishing the retention and binding of the battery cells with a single battery module. In contrast, technology has been of focus which forms a battery pack by abolishing or simplifying the battery module and

mounting a plurality of cell groups consisting of a plurality of battery cells as is into the housing. At this time, the retaining and binding of the battery cells is first accomplished by the cell groups being assembled in the housing. Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2019-106264

## SUMMARY OF THE INVENTION

(5) For such a battery pack, there is room for further improvement, and thus the present inventors have focused on the point that, if able to efficiently fix the cell group to the housing, the production of the battery pack is accelerated, and the quality of the battery pack improves.

(6) The present invention has been made taking account of the above situation, and has an object of configuring so as to be able to efficiently fix a cell group to a housing.

(7) The present inventors found that it is possible to efficiently fix a cell group to a housing so long as providing an injection channel for injecting paste to a partition wall of a housing, and injecting the paste from this injection channel to a gap between the partition wall and the cell group, thereby arriving at the present invention. The present invention is a battery pack according to the following first to fifth aspects.

(8) A battery pack according to a first aspect of the present invention includes: a housing having inside thereof a partition wall extending in an extending direction and projecting in a projecting direction which is orthogonal to the extending direction; a cell group having side-by-side in the extending direction a plurality of battery cells on a side of the partition wall; and paste filled into a gap between the partition wall and the cell group, wherein the partition wall includes an injection channel having an injection port for injecting the paste, and an inlet which communicates with the injection port and opens to the gap.

(9) According to the first aspect of the present invention, by injecting paste to the injection port of the partition wall, and extracting paste from the inlet to the gap between the partition wall and cell group, it is possible to efficiently pour paste into this gap. It is thereby possible to efficiently fix the cell group to the housing.

(10) According to a second aspect of the present invention, in the battery pack as described in the first aspect, the injection port is provided at an end face on a side in the projecting direction of the partition wall, and the inlet is provided more to an opposite direction side of the projecting direction than a center in the projecting direction of a wall face of the partition wall on a side of the cell group.

(11) According to the second aspect of the present invention, the injection port is at an end face of the partition wall on a side in the projecting direction, the access from injection equipment is favorable. Moreover, since the inlet is more to an opposite direction side of the projecting direction than the center in the projecting direction of the partition wall, it is possible to actively lead the paste to the back of the gap. For this reason, it is possible to make the paste uniform by efficiently evacuating the air inside the gap.

(12) According to a third aspect of the present invention, the battery pack as described in the first or second aspect further includes a plate extending in the extending direction along a lateral face of the cell group, in which the plate includes a flange abutting an end face of a plurality of the battery cells on a side in the projecting direction, and a fixed part which is fixed to an end face of the partition wall on a side in the projecting direction.

(13) According to the third aspect of the present invention, by the flange abutting the end face of the plurality of battery cell on a side in the projecting direction, and the fixed part being fixed to the partition wall, it is possible to position and retain the plurality of battery cells in the projecting direction.

(14) According to a fourth aspect of the present invention, in the battery pack as described in the third aspect, the plate includes at intervals in the extending direction a plurality of protrusions projecting to a side of the partition wall within the gap.

(15) According to the fourth aspect of the present invention, by the plurality of protrusions

provided side-by-side in the extending direction, it is possible to secure the gap between the plate and wall surface and it becomes easy to arrange the cell group at the desired position in a lateral direction. In this state, by pouring the paste between the partition wall and cell group, the cell assembly tends to be positioned and retained at the desired position in a lateral direction.

(16) According to a fifth aspect of the present invention, in the battery pack as described in the third aspect, the plate connects the battery cells by being attached to a lateral face of the plurality of the battery cells.

(17) According to the fifth aspect of the present invention, since the battery cells are connected by the plate, it is possible to efficiently assemble the cell group to the housing compared to a case of not connecting thereby. Moreover, this connection simply attaches the plate to a lateral face of a plurality of battery cells, it can be carried out easily, and leads to an improvement in productivity of the battery pack. Furthermore, after assembly of the cell group to the housing, since the cell group is fixed to the housing by paste or the like, there is no particular problem even if the plate distances from the plurality of battery cells. For this reason, this assembly is simple, and this point leads to an improvement in the productivity of the battery pack.

(18) According to the present invention, it is possible to efficiently fix a cell group to a housing.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view showing a battery pack of a first embodiment;

(2) FIG. 2 is a perspective view showing a cell assembly;

(3) FIG. 3 is a perspective view showing an aspect upon assembling the cell assembly in a housing;

(4) FIG. 4 is a perspective view showing a state in which the cell assembly is assembled in the housing;

(5) FIG. 5 is a cross-sectional view showing a cross section along a line V-V in FIG. 4;

(6) FIG. 6 is a plan view showing an electrical connection portion of adjacent cell assemblies;

(7) FIG. 7 is a perspective view showing a second terminal block;

(8) FIG. 8 is an exploded perspective view showing an electrical connection portion of adjacent cell assemblies;

(9) FIG. 9 is a perspective view showing the same connection portion;

(10) FIG. 10 is a perspective view showing a bus bar retention member;

(11) FIG. 11 is an exploded perspective view showing a bus bar retention member and terminal block;

(12) FIG. 12 is a perspective view of a partial cross section showing a bus bar retention member and terminal block; and

(13) FIG. 13 is a perspective view taking a cross section of a different portion than FIG. 12.

### DETAILED DESCRIPTION OF THE INVENTION

(14) Hereinafter, embodiments of the present invention will be explained while referencing the drawings. However, the present invention is not to be limited in any way to the following embodiments, and can be realized by appropriately modifying within a scope not departing from the gist of the present invention.

#### First Embodiment

(15) FIG. 1 is a perspective view showing a battery pack **100** of the present embodiment. The battery pack **100** has a housing **80**, and a plurality of cell assemblies **40** equipped to the housing **80**.

(16) Hereinafter, the predetermined three directions which are orthogonal to each other are respectively referred to as “front/rear direction X”, “left/right direction Y” and “up/down direction Z”. Consequently, the front/rear direction X referred to herein does not necessary indicate the vehicle longitudinal direction, i.e. front/rear direction of the vehicle. More specifically, the

front/rear direction X may be the vehicle longitudinal direction, or may be the vehicle width direction, for example. It should be noted that “front/rear direction X”, “left/right direction Y” and “up/down direction Z” may be replaced with “extending direction”, “lateral direction” and “projecting direction”, respectively.

(17) The housing **80** has a rectangular housing main body **80a** which opens upwards, and a lid (not illustrated) which seals this opening of the housing main body **80a**. The inner side of the housing **80** is divided into a plurality of installation spaces S, by a partition wall **88** projecting upwards from the inner bottom face of the housing main body **80a** and extending in the left/right direction Y, and a plurality of partition walls **85** projecting upwards from the inner bottom face of the housing main body **80a** and extending in the front/rear direction X. One cell assembly **40** is equipped into each installation space S.

(18) FIG. 2 is a perspective view showing the cell assembly **40**. The cell assembly **40** has a cell group **30G**, first plate **10** and second plate **20**. The cell group **30G** has a plurality of battery cells **30** side-by-side in the front/rear direction X. The first plate **10** and second plate **20** extend in the front/rear direction X. The first plate **10** connects the right-side faces of the battery cells **30**, by being attached to the right-side face of each battery cell **30** of the cell assembly **40** to which it belongs by way of adhesive, double-sided top or the like. The second plate **20** connects left-side faces of the battery cells **30** by being attached to the left-side face of each battery cell **30** of the cell assembly **40** to which it belongs by way of adhesive, double-sided tape or the like. As the materials of the first plate **10** and second plate **20**, a high strength resin possessing both an insulation property and mechanical strength is desirable.

(19) The first plate **10** has a first flange **13**, a first fixed part **18**, a plurality of first protrusions **15**, and a first terminal block **12**. This first flange **13**, first fixed part **18**, first protrusion **15** and first terminal block **12** may be formed separately from the main body portion of the first plate **10**, and assembled to the main body portions; however, they are preferably integrally formed with the main body portion.

(20) The first flange **13** projects from the upper end of the first plate **10** laterally inwards, i.e. leftwards yL, and abuts the upper face of the cell group **30G**. The first fixed part **18** projects from the right-side face upper part of the first plate **10** to laterally outwards, i.e. rightwards yR, and a bolt through-hole **18h** piercing vertically is provided.

(21) The first protrusion **15** is a protrusion for securing a gap g between the first plate **10** and a partition wall **85** adjacent thereto, and is arranged in parallel providing a space in the front/rear direction X at the right-side face upper part of the first plate **10**. Each first protrusion **15** projects laterally outwards, i.e. rightwards yR, within this gap g. The first terminal block **12** protrudes from the upper part of the first plate **10** to laterally outwards, i.e. rightwards yR.

(22) The second plate **20** has a second flange **23**, a second fixed part **28**, a plurality of second protrusions **25** and a second terminal block **21**. The explanation for the second plate **20** is read by substituting terms in the explanation of the first plate **10** as follows. Namely, “first” is substituted for “second, and each of “right” and “left” is substituted with the other, and the reference number is substituted with the corresponding reference number. However, the first fixed part **18** and second fixed part **28** are provided at positions shifted in the front/rear direction X from each other, and similarly, the first terminal block **11** and second terminal block **21** are provided at positions shifted in the front/rear direction X from each other.

(23) The adhesive strength of each plate **10**, **20** to the cell group **30G** is sufficient so long as having even the minimum strength and reliability that can withstand the subsequent handling in the assembly work of the cell assembly **40** to the housing **80**. In addition, prior to the cell assembly **40** being assembled to the housing **80**, it shall be handled with each battery cell **30** at low SOC, and thus charging is not carried out. This is because, by each battery cell **30** expanding from charging, there is concern over the adhesion of each plate **10**, **20** to the cell group **30G** peeling away. It should be noted that SOC is an abbreviation of “state of charge”, and indicates the charge state of

the battery, i.e. percentage of remaining capacity accounting for the capacity of the battery.

(24) The cell assembly **40** further includes a plurality inter-cell bus bars **33**, a first bus bar **31** and a second bus bar **32**. The terminals p, n of the battery cell **30** are electrically connected by the inter-cell bus bar **33**. More specifically, in the present embodiment, the plurality of battery cells **30** constituting the cell group **30G** are connected **2** in parallel, i.e. series in two rows, by the plurality of inter-cell bus bars **33**.

(25) The negative electrode terminal n of the battery cell **30** on the lowest potential side constitutes the negative electrode terminal Gn of the overall cell group **30G**. On the other hand, the positive electrode terminal p of the battery cell **30** on the highest potential side constitutes the positive electrode terminal Gp of the overall cell group **30G**. The first bus bar **31** has one end electrically connected to the negative electrode terminal Gn of the overall cell group **30G**, and the other end arranged within the first terminal block **12**. The second bus bar **32** has one end electrically connected to the positive electrode terminal Gp of the overall cell group **30G**, and the other end inserted into the second terminal block **21**.

(26) These bus bars **31** to **33** are mounted after adhesion of the plates **10**, **20** to the cell group **30G**. Furthermore, when a voltage detection line, temperature detection line, etc. omitted from the drawings are attached to the cell group **30G**, and the insulation cover omitted from the drawings is attached to the upper face, etc. of the cell group **30G**, the cell assembly **40** is completed.

(27) FIG. **3** is a perspective view showing an aspect upon mounting the cell assembly **40** in the installation space S. In a portion positioned below the installation space S of the housing **80**, a coolant pipe omitted from the drawings is inlaid to cool the cell group **30G** from the underside. Thermal conduction paste Tp is made to cover the inner bottom face of each installation space S in advance. However, it may be configured so as to install a heat transfer sheet in place of the thermal conduction paste Tp.

(28) FIG. **4** is a perspective view showing a state in which the cell assembly **40** is mounted in a installation space S. Next, positioning and retention in each of the directions X, Y, Z of the cell assembly **40** will be explained.

(29) First, positioning and retention in the front/rear direction X will be explained. After inserting the cell assembly **40** in the installation space S, a shim **39** is inserted between one end in the front/rear direction X of the cell assembly **40** and the inner wall of the installation space S. At this time, the appropriate thickness shim **39** is selected so that the load in the front/rear direction X against the cell assembly **40** after insertion of the shim **39** becomes suitable, i.e. 1 to 2 kN, for example. The positioning and retention in the front/rear direction X of the cell assembly **40** thereby completes. It should be noted that it is desired to be a high strength resin possessing both strength and insulation property as the material of the shim **39**.

(30) Next, positioning in the up/down direction Z of the cell assembly **40** will be explained. The bolt B is made to insert into the bolt through hole **18h**, **28h** of the fixed parts **18**, **28** of each plate **10**, **20**, and this bolt B is threaded in the bolt holes **81**, **82** which is in the upper face of the partition wall **85**, whereby the fixed parts **18**, **28** are fastened to the upper face of the partition wall **85**. Positioning and retention in the up/down direction Z of the cell assembly **40** are thereby completed.

(31) In other words, the cell group **30G** is retained in a downwards undisplaceable manner, by abutting to the inner bottom face of the load space S via the thermal conduction paste Tp on the lower face of the cell group **30G**. Then, by abutting of the flanges **13**, **23** to the upper face of the cell group **30G** and fastening of the fixed parts **18**, **28** to the upper face of the partition wall **85**, the cell group **30G** is retained in an upward undisplaceable manner. The cell assembly **40** is thereby positioned and retained in the up/down direction Z. It should be noted that, at this time, part of the thermal conduction paste Tp coated in advance on the bottom face of the installation space S goes around part of the gap g between the cell assembly **40** and partition wall **85**.

(32) FIG. **5** is a cross-sectional view showing a cross section along the line V-V in FIG. **4**. Next, positioning in the left/right direction Y of the cell assembly **40** will be explained. As described

above, the first plate **10** and second plate **20** include the protrusions **15**, **25** for securing the gap **g** from each partition wall **85**. For this reason, upon mounting the cell assembly **40** in the installation space **S**, the cell assembly **40** is centered in the left/right direction **Y** by the protrusions **15**, **25** on both left and right sides. It should be noted that, due to the dimensional tolerance, etc. of the housing main body **80a**, the protrusions **15**, **25** can be crushed or shaved off during mounting of the cell assembly **40**; however, these protrusions **15**, **25** have an object of only centering of the cell assembly **40**, and do not necessarily assume a retaining function of the cell assembly **40**; therefore, there is no particular problem.

(33) The partition wall **85** has an injection channel **90** for injecting the paste **P** into the gap **g** between the partition wall **85** and cell assembly **40**. The paste **P** is resin or adhesive of a type having fluidity during injection, and which subsequently hardens irreversibly, or is a paste like these. The injection channel **90** has a vertical extension **92** which extends from an injection port **91** provided at the upper face of the partition wall **85** towards downwards, and a left/right extension **98** extending from a lower end of the vertical extension **92** to the left/right direction. The left end of the left/right extension **98** configures an inlet **99** on the left side opening at the left-side face of the partition wall **85**, and a right end of the left/right extension **98** configures an inlet **99** on the right side opening at the right-side face of the partition wall **85**. Each of the left/right inlets **99** communicates with the injection port **91**, and opens to the gap **g** in the lower part or lower end of the partition wall **85**.

(34) During assembly of the cell assembly **40** to the housing **80**, paste **P** is injected from the injection channel **90** to the gap **g** between the plates **10**, **20** and partition wall **85**. When this paste **P** hardens, positioning and retention in the left/right direction **Y** of the cell assembly **40** completes. It should be noted that the paste **P** has an object of retention of the cell assembly **40**, and thus it is not a problem even if the paste **P** not particularly having thermal conductivity. In addition, the paste **P** does not need to fully fill the gap **g**, and is sufficient if filling at least the lower part of the gap **g**. This is because the upper parts of the plates **10**, **20** are positioned and retained not only in the up/down direction **Z**, but also the left/right direction **Y**, by the fastening of the fixed parts **18**, **28** relative to the upper face of the partition wall **85**.

(35) As above, the cell assembly **40** is positioned and retained in the respective directions **X**, **Y**, **Z** relative to the housing **80**. It should be noted that adhesion of the battery cells **30** by the plates **10**, **20** also contributes to retaining of the cell group **30G**; however, it is not a problem even if adhesion is peeled apart after assembly of the cell assembly **40** to the housing **80**. In addition, it is not a problem even if adhesive strength or adhesive force of the paste **P** is lost. This is because, as long as the solidified paste **P** still exists in the gap **g**, the positioning and retaining in the left/right direction **Y** of the cell assembly **40** will function.

(36) FIG. **6** is a plan view from above of an electrical connection portion of adjacent cell assemblies **40** in the left/right direction **Y** and the vicinity thereof. The first terminal block **12** on the left-side cell assembly **40** and the second terminal block **21** on the right-side cell assembly **40** project directly above the same partition wall **85**. A coupling bus bar **50** has one end inserted in the first terminal block **12** of the left-side cell assembly **40**, and the other end inserted in the second terminal block **21** of the right-side cell assembly **40**, whereby the left-side cell assembly **40** and right-side cell assembly **40** are electrically connected within the housing **80** and directly above the partition wall **85**. In addition, stated differently, the coupling bus bar **50** electrically connects a “first cell group” as the cell group **30G** of the left-side cell assembly **40**, and a “second cell group” as the cell group **30G** of the right-side cell assembly **40**.

(37) FIG. **7** is a perspective view of the second terminal block **21** of the right-side cell assembly **40** viewed obliquely from above. The second terminal block **21** has an opening **21a**. The opening **21a** opens at least to the side of the first terminal block **12** and upwards. The U-shaped second connection member **21b** in a front view seen in the front/rear direction **X** is provided inside of the second terminal block **21**. The second connection member **21b** is electrically connected to the

second bus bar **32**, by abutting the second bus bar **32**.

(38) FIG. **8** is an exploded perspective view seeing the electrical connection portion of adjacent cell assemblies **40** in the left/right direction Y and the vicinity thereof from obliquely above on the front/rear opposite side than the case of FIG. **7**.

(39) The first terminal block **12** has an opening **12a**, and the first connection member **12b** is installed thereto. Explanation of the first terminal block **12** is read by substituting terms in the explanation of the second terminal block **21** as follows. Namely, each one of “first” and “second” is substituted with the other, and the reference number is substituted with a corresponding one.

(40) The coupling bus bar **50** is a member extending in the front/rear direction X, and an insulation cover **55** is fit onto a central part in the length direction. The coupling bus bar **50** is inserted inside of the first terminal block **12** and second terminal block **21**, from the upper side of the openings **12a**, **21a** of the first terminal block **12** and second terminal block **21**.

(41) FIG. **9** is a perspective view showing a state inserting the coupling bus bar **50** inside of the first terminal block **12** and second terminal block **21**. By one end abutting the first connection member **12b** and the other end abutting the second connection member **21b**, the coupling bus bar **50** electrically connects the first bus bar **31** of the cell assembly **40** on the leftward yL side, i.e. right side in FIG. **9**, and the second bus bar **32** of the cell assembly **40** on the rightward yR side, i.e. left side in FIG. **9**. The coupling bus bar **50** thereby electrically connects the leftward yL-side cell assembly **40** and the rightward yR-side cell assembly **40** within the housing **80** and directly above the partition wall **85**.

(42) To each terminal block **12**, **21**, a retainer **60** serving as a bus bar retention member which is an insulator is attached combining insulation of the coupling bus bar **50** relative to the periphery of these terminal blocks **12**, **21**, and fall prevention of the coupling bus bar **50**.

(43) FIG. **10** is a perspective view showing the retainer **60**. The retainer **60** has a horizontal part **61** projecting inwards left and right, i.e. to the cell group **30G** side, and a vertical part **65** projecting downwards from the outward ends on left/right of the horizontal part **61**. At the lower face of the horizontal part **61**, ribs **62** projecting downwards are arranged side-by-side at an interval in the front/rear direction X. In the vertical part **65**, penetrating hole-like engagement holes **66** are arranged side-by-side at an interval in the front/rear direction X.

(44) FIG. **11** is an exploded perspective view showing the retainer **60** and periphery thereof. The terminal block **21** has snap fits **21c** engaging the engagement holes **66** of the retainer **60** at a lateral face at an interval in the front/rear direction.

(45) FIG. **12** is a perspective view of a partial cross section showing a state in which the retainer **60** is attached to the terminal block **21**, without the horizontal part **61**. An anterior rib **62** abuts the inside face of a front side and inside faces on the left/right of the terminal block **21**, and a posterior rib **62** abuts the inside face on a rear side and inside faces on the left/right of the terminal block **21**, whereby the horizontal part **61** engages the upper end of the terminal block **21**. In addition, together with this, by the snap fits **21c** engaging each engagement hole **66**, the vertical part **65** engages the outward lateral faces on the left/right of the terminal block **21**.

(46) FIG. **13** is a perspective view of a partial cross section showing a state in which the retainer **60** is attached to the terminal block **21** without the left/right outward side of the retainer **60** and terminal block **21**. The horizontal part **61** insulates the coupling bus bar **50** from the periphery of the terminal block **21**, by covering the end in the extending direction of the coupling bus bar **50** from above. The rib **62** prevents falling off of the coupling bus bar **50**, by abutting from the upper side the coupling bus bar **50**.

(47) According to the present embodiment shown above, it is possible to improve the productivity and quality of the battery pack **100**, and improve the mounting efficiency of the cell assembly **40** to the housing **80**. The details thereof will be explained below.

(48) By attaching the plates **10**, **20** to a lateral face of each cell group **30G**, the plurality of battery cells **30** are connected by the plates **10**, **20**. For this reason, compared to a case not connected, it is



possible to efficiently perform assembly of the cell group **30G** to the housing **80**. Moreover, due to being assembly by adhesive, double-sided tape or the like, it is possible to carry out easily.

(49) Furthermore, due to interposing the shim **39** between one end in the front/rear direction X of the cell assembly **40** and the inner wall of the installation space S, it is possible to position and retain the cell assembly **40** in the front/rear direction X by the shim **39**.

(50) Moreover, by having the flanges **13**, **23** of the plates **10**, **20** abut the upper face of the cell group **30G**, and fastening the fixed parts **18**, **28** of the plates **10**, **20** to the upper face of the partition wall **85**, it is possible to position and retain the cell assembly **40** in the up/down direction Z.

(51) Furthermore, due to having the protrusions **15**, **25** projecting left/right outwards at the left and right plates **10**, **20** respectively, upon inserting the cell assembly **40** in the installation space S, it is possible to secure the gap g between the plates **10**, **20** and the partition wall **85** by these protrusions **15**, **25**, and possible to center the cell group **30G** in the left/right direction Y. In this state, by pouring the paste P in the gap g between the partition wall **85** and cell group **30G**, it is possible to prevent a situation, for example, in which the gap g more to one side in the left/right direction than the cell group **30G** is filled with the paste P with a large thickness in the left and right direction Y at a low density, and the gap g more to the other side in the left/right direction than the cell group **30G** is filled with the paste P with a small thickness in the left and right direction Y at a high density. For this reason, it is possible to position and retain the cell assembly **40** with good balance in the left/right direction Y.

(52) Moreover, since the injection channel **90** is provided in the partition wall **85**, it is possible to efficiently pour the paste P in the gap g between the partition wall **85** and cell group **30G**. Furthermore, since the injection port **91** is in the upper face of the partition wall **85**, access from the injection equipment is favorable. Moreover, upon attaching a lid to the housing main body **80a**, since the injection port **91** is covered by this lid, protection of the injection port **91** is also favorable. Additionally, since the inlet **99** is in the lower part or lower end of the partition wall **85**, it is possible to actively lead the paste P to the back of the gap g. For this reason, it is possible to make the paste P uniform by efficiently evacuating the air inside the gap g.

(53) Moreover, since the cell assembly **40** is positioned and retained in the respective directions X, Y, Z in the above way, after assembly of the cell assembly **40** to the housing **80**, there is no particular problem even if adhesion of the plates **10**, **20** to the cell group **30G** peels away. For this reason, this adhesion is sufficient even if simple adhesion.

(54) Furthermore, each cell assembly **40** is not a so-called “end plate” at both ends in the front/rear direction X. For this reason, compared to a case of providing a terminal block at the end plate, it is possible to make the cell assembly **40** compact in the front/rear direction X. It should be noted that the end plate is a plate-like member positioned at both ends in the front/rear direction X of the cell group **30G** in a conventional battery module. By the end plates on both ends being connected in the front/rear direction X by a metallic bind bar, force in the front/rear direction X is applied to the cell group **30G** and the cell group **30G** is retained.

(55) In addition, in the present embodiment, the first terminal block **12** of the left-side cell assembly **40** and the second terminal block **21** of the right-side cell assembly **40** within the housing **80** project directly above the same partition wall **85**, and the coupling bus bar **50** electrically connects the first bus bar **31** of this first terminal block **12** and the second bus bar **32** of this second terminal block **21**. For this reason, effectively using the space within the housing **80** and directly above the partition wall **85**, it is possible to electrically connect adjacent cell groups **30G** in the left/right direction Y.

(56) Furthermore, at this time, since the first terminal block **12** and second terminal block **21** are shifted in the front/rear direction X, it is possible to avoid interference between the first terminal block **12** and second terminal block **21**, and make adjacent cell assemblies **40** in the left/right direction Y to approach more in the left/right direction Y. For this reason, it is possible to thin the partition wall **85**, and possible to pack and arrange the plurality of cell assemblies **40** in the

left/right direction Y.

(57) Moreover, since the terminal blocks **12, 21** are formed in the plates **10, 20**, it is possible to reduce the number of parts compared to the case of the terminal blocks **12, 21** being separate members than the plates **10, 20**.

(58) In addition, each terminal block **12, 21** has an opening **21a** which opens upwards and towards the partnering terminal block **21, 12**. For this reason, by simply plugging in the coupling bus bar **50** to both terminal blocks **12, 21** from the upper side of both openings **12a, 21a**, it is possible to simply assemble the coupling bus bar **50** to both terminal blocks **12, 21**.

(59) Furthermore, the retainer **60** of the insulator covering the openings **12a, 21a** is attached to the terminal blocks **12, 21**. For this reason, it is possible to insulate the coupling bus bar **50** from the periphery of the terminal blocks **12, 21** by the retainer **60**, and prevent falling out of the coupling bus bar **50**.

(60) In the above way, it is possible to improve the productivity and quality of the battery pack **100**, and improve the mounting efficiency of the cell assembly **40** to the housing **80**.

#### Other Embodiments

(61) The embodiments shown above can be modified as follows, for example. In the case that there is no particular problem in insulation and retention of the coupling bus bar **50** even when installing the retainer **60** to only one among the first terminal block **12** and second terminal block **21**, it may be configured so as to install the retainer **60** to only this one.

#### EXPLANATION OF REFERENCE NUMERALS

(62) **10** first plate **12** first terminal block **12a** opening **13** first flange **15** first protrusion **18** first fixed part **20** second plate **21** second terminal block **21a** opening **23** second flange **25** second protrusion **28** second fixed part **30** battery cell **30G** cell group **31** first bus bar **32** second bus bar **40** cell assembly **50** coupling bus bar **60** retainer as bus bar retention member **80** housing **90** injection channel **91** injection port **99** inlet **100** battery pack P paste

## Claims

1. A battery pack comprising: a housing having inside thereof a partition wall extending in an extending direction and projecting in a projecting direction which is orthogonal to the extending direction; a cell group having side-by-side in the extending direction a plurality of battery cells on a side of the partition wall; paste filled into a gap between the partition wall and the cell group; and a plate extending in the extending direction along a lateral face of the cell group, wherein the partition wall includes an injection channel having an injection port for injecting the paste, and an inlet which communicates with the injection port and opens to the gap, and wherein the plate includes a flange abutting an end face of a plurality of the battery cells on a side in the projecting direction, and a fixed part which is fixed to an end face of the partition wall on a side in the projecting direction.
  2. The battery pack according to claim 1, wherein the injection port is provided at an end face on a side in the projecting direction of the partition wall, and the inlet is provided more to an opposite direction side of the projecting direction than a center in the projecting direction of a wall face of the partition wall on a side of the cell group.
  3. The battery pack according to claim 1, wherein the plate includes at intervals in the extending direction a plurality of protrusions projecting to a side of the partition wall within the gap.
  4. The battery pack according to claim 1, wherein the plate connects the battery cells by being attached to a lateral face of the plurality of the battery cells.
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